In the following by ultrafilter on X we mean an ultrafilter on $P(X)$. Denote τ_x the open neighbourouds of x. We say $U \in St(X)$ converges to $x \in X$ meaning that every open neighbouroud of x is in U.

Esercizio 1. *Let (X, τ) be a topological space. Then tfae:*

1. *X compact*
2. *$\forall U \in St(X)$ U converges to some $x \in X$*

Proof.

- 1) $\Rightarrow$ 2): By contradiction assume

$$\exists U \in St(X) \forall x \in X \exists V_x \in \tau_x V_x \notin U$$

Let $C_x = X \setminus V_x$. Then $\forall x \in X$ $C_x \in U$. Observe that $\{V_x : x \in X\}$ is an open covering of X, so by compactness we can extract a finite sub-covering $\{V_{x_1}, .., V_{x_n}\}$. Then $\bigcap_{i=1,..,n} C_{x_i} = \emptyset$, contradiction since $\emptyset \notin U$.

- 2) $\Rightarrow$ 1): By contradiction assume $\{U_i : i \in I\}$ is some covering of X that does not admit a finite sub-covering. Denote $V_i := X \setminus U_i$ for all $i \in I$. This gives that $\{V_i : i \in I\}$ are closed subsets of X with the fip. Then there exists some ultrafilter U such that $\{V_i : i \in I\} \subseteq U$. Let $x \in X$ be convergence point for U. Since $\{U_i : i \in I\}$ is a covering of X there must be some j such that $x \in U_j$. But then $U_j \in U, V_j \in U$, which implies $\emptyset \in U$, contradiction.

□

Now we state a generalization on the condensation lemma. Given X topological space and $(x_i : i \in I) \subseteq X$ with I infinite, let U be ultrafilter on I. We say that x is limit along U of $(x_i : i \in I)$ if

$$\forall V \in \tau_x \{i \in I : x_i \in V\} \in U$$

If the limit is unique (this is the case if τ is $T2$) we will denote $x = lim_U(x_i : i \in I)$.

Esercizio 2. *Let (X, τ) compact, $T2$ topological space. Let $(x_i : i \in I) \subseteq X$, with $|X| \geq \omega$. Let U ultrafilter on I. Then there exists unique limit along U of $(x_i : i \in I)$.*

Proof. Consider $f : I \to X, i \mapsto x_i$. We can define $f_* : St(I) \to St(X), H \mapsto \{A \subseteq X : f^{-1}(A) \in H\}$. It is easy to verify that f_* is well defined. Now $f_*(U)$ is an ultrafilter on X, compact, so we can apply the exercise above to see that there is a limit for $(x_i : i \in I)$ along $f_*(U)$. This is unique since X T_2. Call x such limit. Then $\forall V \in \tau_x V \in f_*(U)$ and by definition of f_* we have $\forall V \in \tau_x \{i \in I : x_i \in V\} \in U$. Equivalently x is limit along U of $(x_i : i \in I)$.

□

Note that in the case $I = \omega$ and $U = \{A \subseteq I : |A^c| < \omega\}$ we obtain that $\exists x \in X \forall V \in \tau_x \exists A \in U\{x_i : i \in A\} \subseteq V$. If X is a metric space, take $V_2 = B(x, 1/2), .., V_n = B(x, 1/n), ..$ and $A_2, .., A_n, ..$ the corresponding element in U Now choose a_n as the minimum element in A_n such that $a_n > a_{n-1}$. Then $\{a_n : n < \omega\}$ gives the thesis of condensation lemma.

Esercizio 3. *G is compact.*

Proof. Fixing an embedding of J in $S_x(M)$ one can assume wlog that $J \subseteq S_x(M)$. Denote $E = End(S_x(M)) = \{f \in S_x(M)^{S_x(M)} : f \text{ endomorphism of } S_x(M)\}$. Let $G = (g_i : i \in I)$ enumeration and let U ultrafilter on I. By the first above exercise it is enough proving that the limit of $(g_i : i \in I)$ along U exists. Use homogeneity of $S_x(M)$ to extend each g_i to $\overline{g_i} \in E$. Equip $S_x(M)^{S_x(M)}$ with the product standard topology. Then $S_x(M)^{S_x(M)}$ is compact and E is a closed subspace. In fact, enumerate $S_x(M) = (p_\alpha : \alpha < \lambda)$ and observe that (by finite character of morphisms)

$$\begin{aligned}((p_\alpha, q_\alpha) : \alpha < \lambda) \in E \\ &\iff \forall n < \omega \forall \alpha_1 < \alpha_2 < .. < \alpha_n loc(p_{\alpha_1}, .., p_{\alpha_n}) \supseteq loc(q_{\alpha_1}, .., q_{\alpha_n}) \\ &\iff \forall n < \omega \forall \alpha_1 < \alpha_2 < .. < \alpha_n ((p_{\alpha_1}, .., p_{\alpha_n}), (q_{\alpha_1}, .., q_{\alpha_n})) \in loc(p_{\alpha_1}, .., p_{\alpha_n})^2\end{aligned}$$

Since $loc(p_{\alpha_1}, .., p_{\alpha_n})$ is closed in $S_x(M)^n$ we get E closed. Now the limit of $(\overline{g_i} : i \in I)$ along U exists unique in $S_x(M)^{S_x(M)}$ since it is compact and T_2. Denote $\overline{g}_*$ such limit. We claim $\overline{g}_* \in E$. By contradiction $\overline{g}_* \in E^c$. Since E^c open and by definition of limit, $A := \{i \in I : \overline{g_i} \in E^c\} \in U$. In particular $A \neq \emptyset$. Let $i \in A$. Then $\overline{g_i} \notin E$. This is a contradiction. Let $f : S_x(M) \to J$ retraction and let $g := (f \circ \overline{g_*}) \restriction_J \in End(J) = Aut(J)$. We claim that g is the limit of $(g_i : i \in I)$ along U. By contradiction, let R, a, b such that $g \in O_{R,a,b} \notin U$. Now observe that

$$\begin{aligned}g \in O_{R,a,b} &\iff J \models \neg R(g(a), b) \\ &\iff J \models \neg R(f(\overline{g}_*(a)), f(b)) \\ &\iff S_x(M) \models \neg R(\overline{g}_*(a), (b)) \\ &\iff \overline{g}_* \in O_{R,a,b}\end{aligned}$$

Since $\overline{g}_*$ is the limit of $(\overline{g_i} : i \in I)$ along U and $O_{R,a,b}$ open in $S_x(M)^{S_x(M)}$, we have $\{i \in I : \overline{g_i} \in O_{R,a,b}\} \in U$. Equivalently $\{i \in I : S_x(M) \models \neg R(\overline{g_i}(a), b)\} \in U$. Since $a, b \in J$, this is the same as $\{i \in I : J \models \neg R(g_i(a), b)\} \in U$. This is a contradiction since $O_{R,a,b}$ should witness g non limit.

□